

JAN-PAUL VINCENT RAMOS-DÁVILA

PERSONAL DATA

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EXPERIENCES

- 2026 - **Network Systems Software Engineer, Hewlett-Packard Enterprise**
Network Test Engineer (II); Design and execute automated system test plans for HPE ArubaOS-CX campus and branch switches within the EDSW team, validating enterprise Layer 2/Layer 3 protocols, Dynamic Segmentation, and high-density traffic performance.
Develop scalable Python and PyTest automation frameworks integrated into CI/CD pipelines to isolate firmware defects, capture packet traces, and drive critical software updates across Aruba switching platforms.
Supervisor: Solution Test and Auto ENG (Sanjeev Kumar)
- 2024 - 2025 **Research Intern, NASA Langley Formal Methods**
Mechanized proofs that model correct behaviors of a Software Defined Delay-Tolerant Network's Match-Action pipeline for NASA's Interplanetary Overlay Network framework.
Developed a formally verified Network Calculus IR in Rocq. Wrote an interpreter for a subset of P4 to target the IR.
Advisor: Dr. Alwyn Goodloe
- 2022 - 2024 **Research Intern, Carnegie Mellon University S3D**
Core contributor on the early development of the [Gradual Verification framework](#). Empirically evaluated the soundness of Gradual C_0 , and provided formal proofs of completeness between the dynamic and static verifiers.
Explored the application of Gradual Verification to smart contracts on the Algorand and Ethereum blockchain platforms and developed a prototype for [Gradually Verified Teal](#).
Advisor: Dr. Jonathan Aldrich
- 2022 - 2024 **Research Assistant, Cornell University CAPRA**
Worked on a symbolic execution tool for verifying parallelism in Calyx.
Advisor: Dr. Adrian Sampson

EDUCATION

- 2025 - ON PAUSE **Boston University**
DOCTOR OF PHILOSOPHY IN COMPUTER SCIENCE
Interests: Programming Languages, Type Theory, Formal Methods
Advisors: Dr. Ankush Das, Dr. Marco Gaboardi
- 2021 - 2025 **Cornell University**
BACHELOR'S OF ARTS IN PHILOSOPHY
Interests: Foundations of Mathematics and Logics, Analytics
Advisors: Dr. Adrian Sampson (CompSci), Dr. Harold Hodes (Phil)
- SUMMERS | [UC/EasyUC'25](#), [SPLV'25](#), [OPLSS'24](#), [AFP'23](#)

PUBLICATIONS

- 2025 | Jenna DiVincenzo, Ian McCormack, Conrad Zimmerman, Hemant Gouni, Jacob Gorenburg, **Jan-Paul Ramos-Dávila**, Mona Zhang, Joshua Sunshine, Éric Tanter, Jonathan Aldrich. “*Gradual Co: Symbolic Execution for Gradual Verification*”, In **TOPLAS**, **46(4)**, Article No.: 14 P.1-57 and **POPL 2025**
- 2023 | **Jan-Paul Ramos-Dávila**. “*Evaluation Soundness of a Gradual Verifier with Property Based Testing*”, In **Cornell Undergraduate Research Journal**, **2(1)**, P.17-27 and **POPL 2023 Student Research Competition**.

PRESENTATIONS

- 2025 | “*Sound Default-Typed Scheme*”, In **Scheme and Functional Programming Workshop**.
- “*Formal Verification of a Software Defined Delay-Tolerant Network*”, In **IEEE Workshop on Optimizing Interplanetary Communication Through Network Autonomy and The Eleventh International Workshop on Coq for Programming Languages**.
- 2024 | “*Gradual Verification of Smart Contracts*”, In **Workshop on Principles of Secure Compilation** and **POPL 2024 Student Research Competition**.
- 2023 | “*Optimization of a Gradual Verifier: Lazy evaluation of Iso-recursive Predicates as Equi-recursive at Runtime*”, In **The Midwest PL Summit 2023** and **POPL 2023 Student Research Competition**.

TEACHING

TEACHING ASSISTANT

- 2025 | **CS 4/5111 Practicum in Operating Systems**
Ran coding workshops with hands-on demos building and debugging C applications while teaching the EGOS operating system.
Cornell University
- 2024 | **CS 4114 Systems Programming**
Graded assignments and ran coding workshops with hands-on demos building and debugging C++/Linux applications.
Cornell University
- CS 4/5110 Programming Languages and Logics**
Examination czar in charge of the infrastructure of midterms, graded students’ assignments, and held weekly office hours.
Cornell University

AWARDS

- 2025 **Graduate Fellowship**, Boston University
- Scholarship**, SPLV Summer School at The University of Edinburgh
- 2024 **Scholarship**, Verification Mentoring Workshop at CAV
- 2023 **Intern Fellowship**, Amazon Summer Undergrad Research Experience at CMU REUSE
- Third Place Winner**, Student Research Competition at ACM SIGPLAN POPL
- 2022 **Scholarship**, PLMW at ACM SIGPLAN PLDI
- 2020/1 **Finalist**, Mathematics at Regeneron International Science and Engineering Fair

ACADEMIC SERVICE

- 2027 **Video Co-Chair**, POPL CDMX MX
- 2026 **Video Co-Chair**, PLDI Boulder CO
- Artifact Evaluation Program Committee**, ETAPS Turin IT
- Student Volunteer**, POPL Rennes FR
- 2025 **Video Co-Chair**, ICFP/SPLASH Singapore
- Video Co-Chair**, PLDI Seoul KR
- Video Co-Chair**, POPL Denver CO
- 2024 **Virtualization Chair**, ICFP Milan IT
- Virtualization Chair**, PLDI Copenhagen DK
- 2024 **Video Co-Chair**, POPL London UK
- 2023 **Video Co-Chair**, SPLASH Cascais PT
- Student Volunteer**, ICFP Seattle WA

CERTIFICATES

- IN PROGRESS **(HPE7-A07) HPE Aruba Networking Certified Expert, Campus Access Mobility**
- IN PROGRESS **(HPE7-A06) HPE Aruba Networking Certified Expert, Campus Access Switching**
- IN PROGRESS **(HPE7-A02) HPE Aruba Networking Certified Network Security Professional**
- IN PROGRESS **(HPE7-A01) HPE Aruba Networking Certified Professional, Campus Access**

SKILLS

- ENGLISH Native
- SPANISH Native
- TOOLS Unix, Git, Bash, Neovim, Docker, HTML/CSS
- PLANGS $\LaTeX$, Rocq, OCaml, Scala, Python, Haskell, JS/TS, Java, Go, C/C++, Rust, P4